

研究 开放获取

开发一种用于筛选中西医结合中肝毒性化合物的人工智能方法

赵晨^{1†}、赵梦竹^{1†}、尤良振^{3†}、郑睿¹、蒋音¹、张晓宇¹、邱瑞金¹、孙洋¹、潘海洽¹、何天迈¹、魏旭旭¹、陈智能²、赵晨^{3*} 和尚弘才^{1*}

摘要
背景: 传统中医与西医结合 (TCM-WMC) 增加了复合体的复杂性。
目标: 基于化学结构, 利用人工智能 (AI) 方法开发一种筛选中药-化学药复方 (TCM-WMC) 肝毒性化合物的方法。
方法: 药物性肝损伤 (DILI) 数据收集自公共数据库和已发表的文献。由DILI数据组成的总数据集大约按3:1的比例随机划分为训练集和测试集。采用SGD (随机梯度下降)、kNN (k近邻)、SVM (支持向量机)、NB (朴素贝叶斯)、DT (决策树)、RF (随机森林)、ANN (人工神经网络)、AdaBoost、LR (逻辑回归) 等机器学习模型, 以及一个深度学习模型 (深度置信网络, DBN) 来构建用于筛选肝毒性化合物的模型。
结果: 本研究收集了2035个肝毒性化合物的数据集, 其中1505个化合物作为训练集, 530个化合物作为测试集。结果显示, 随机森林 (RF) 在训练集上获得了0.838的分类准确率 (CA)、0.827的F1分数、0.832的精确率、0.838的召回率、0.814的曲线下面积 (AUC), 在测试集上获得了0.767的CA、0.731的F1、0.739的精确率、0.767的召回率、0.739的AUC, 表现优于其他八种机器学习方法。深度信念网络 (DBN) 在测试集上获得了82.2%的准确率, 高于其他任何机器学习模型在测试集上的表现。
结论: 预计DILI人工智能模型能够有效筛选中药中的肝毒性化合物。
关键词: 人工智能, 药物性肝损伤, 机器学习, 深度学习, 中医-西医结合, 安全性评估 WMC.

† 赵晨、赵梦竹和尤亮珍对本工作贡献相同 *通讯作者:
zyyzc2007@163.com; shanghongcai@126.com

¹ 教育部中医内科学重点实验室, 北京中医药大学东直门医院, 北京, 中国³ 临床医学基础研究所, 中国中医科学院, 北京, 中国 文章末尾提供完整作者信息列表

介绍
药物性肝损伤 (DILI) 是最常见且严重的不良药物反应之一, 是终止临床试验或撤回新药的主要原因 [1]。统计分析显示, 在47种被撤回的药物中, 有15种由于肝毒性, 占31.9% [2]。一项观察性队列研究表明, DILI是急性肝损伤 (ALF) 的主要原因。



© 2022 作者, 已更正出版 2022。开放获取 本文章根据知识共享署名 4.0 国际许可协议授权, 允许以任何媒介或格式使用、共享、改编、分发和复制, 只要您给予原作者和来源适当的署名, 提供知识共享许可链接, 并注明是否进行了修改。除非在材料的署名中另有说明, 本文章中的图像或其他第三方材料均包含在本文章的知识共享许可中。如果材料不包含在本文章的知识共享许可中, 且您的预期使用未被法定条例许可或超出允许使用范围, 则需要直接从版权所有者处获得许可。欲查看该许可副本, 请访问 <http://creativecommons.org/licenses/by/4.0/>。知识共享公共领域奉献豁免 (<http://creativecommons.org/publicdomain/zero/1.0/>)

RESEARCH Open Access

Developing an artificial intelligence method for screening hepatotoxic compounds in traditional Chinese medicine and Western medicine combination

Zhao Chen^{1†}, Mengzhu Zhao^{1†}, Liangzhen You^{3†}, Rui Zheng¹, Yin Jiang¹, Xiaoyu Zhang¹, Ruijin Qiu¹, Yang Sun¹, Haie Pan¹, Tianmai He¹, Xuxu Wei¹, Zhineng Chen², Chen Zhao^{3*} and Hongcai Shang^{1*}

Abstract
Backgrounds: Traditional Chinese medicine and Western medicine combination (TCM-WMC) increased the complexity of compounds ingested.
Objective: To develop a method for screening hepatotoxic compounds in TCM-WMC based on chemical structures using artificial intelligence (AI) methods.
Methods: Drug-induced liver injury (DILI) data was collected from the public databases and published literatures. The total dataset formed by DILI data was randomly divided into training set and test set at a ratio of 3:1 approximately. Machine learning models of SGD (Stochastic Gradient Descent), kNN (k-Nearest Neighbor), SVM (Support Vector Machine), NB (Naive Bayes), DT (Decision Tree), RF (Random Forest), ANN (Artificial Neural Network), AdaBoost, LR (Logistic Regression) and one deep learning model (deep belief network, DBN) were adopted to construct models for screening hepatotoxic compounds.
Result: Dataset of 2035 hepatotoxic compounds was collected in this research, in which 1505 compounds were as training set and 530 compounds were as test set. Results showed that RF obtained 0.838 of classification accuracy (CA), 0.827 of F1-score, 0.832 of Precision, 0.838 of Recall, 0.814 of area under the curve (AUC) on the training set and 0.767 of CA, 0.731 of F1, 0.739 of Precision, 0.767 of Recall, 0.739 of AUC on the test set, which was better than other eight machine learning methods. The DBN obtained 82.2% accuracy on the test set, which was higher than any other machine learning models on the test set.
Conclusion: The DILI AI models were expected to effectively screen hepatotoxic compounds in TCM-WMC.
Keywords: AI, DILI, Machine learning, Deep learning, TCM-WMC, Safety assessment

[†]Zhao Chen, Mengzhu Zhao and Liangzhen You contributed equally to this work

*Correspondence: zyyzc2007@163.com; shanghongcai@126.com

¹ Key Laboratory of Chinese Internal Medicine of Ministry of Education, Dongzhimen Hospital, Beijing University of Chinese Medicine, Beijing, China
³ Institute of Basic Research in Clinical Medicine, China Academy of Chinese Medical Sciences, Beijing, China
Full list of author information is available at the end of the article



© The Author(s) 2022, corrected publication 2022. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>. The Creative Commons Public Domain Dedication waiver (<http://creativecommons.org/publicdomain/zero/1.0/>) applies to the data made available in this article, unless otherwise stated in a credit line to the data.

Introduction
Drug-induced liver injury (DILI), one of the commonest and serious adverse drug reactions, is the dominant cause for terminating clinical trials or withdrawing new drugs [1]. Statistical analysis showed that 15 of the 47 withdrawn drugs were due to hepatotoxicity, accounting for 31.9% [2]. An observational cohort study showed that DILI was the main cause for acute liver injury (ALF),

在美国占 11 1998 2013 %之 [3] 23.8 100 间。中国的DILI发病率为每千人。 [4], 引起了更多的研究关注。

药物性肝损伤 (DILI) 的发病机制尚不完全清楚。许多风险因素可导致DILI, 如宿主的遗传、免疫和代谢因素, 以及药物和环境的影响 [5, 6]。尽管已经开发了许多方法来预测和评估DILI的风险, 但成熟且高精度的方法仍然不足。经典方法, 包括体外和体内实验, 在预测DILI中起着关键作用。然而, 由于多种疾病的共存, 药物联合使用已变得普遍。中药复方 (TCM-WMC) 也越来越受到认可 [7–9], , 几乎涵盖中国所有临床治疗领域 [10], , 这增加了摄入化合物的复杂性。因此, 开发一种有效的方法来筛查中药复方化合物的肝毒性是十分必要的。

随着新型计算技术的发展, 人工智能 (AI) 模型已被广泛应用于化学信息学 [11], 、医学影像 [12], 、诊断学 [13], 、生物信息学 [14], 等领域, 为肝毒性化合物的筛选提供了新的思路。机器学习和深度学习方法已越来越多地应用于肝毒性化合物的筛选, 这些方法将高维化学结构信息视为向量, 并以高效的方式进行预测或分类计算 [15–17]。在本研究中, 收集了DILI数据集以建立 AI 模型, 用于中药-药物复方 (TCM-WMC) 中肝毒性化合物的筛选。

方法 方法设计

DILI数据集是从公共数据库和已发表文献中收集的。结合 DILI 数据集构建了九个机器学习模型和一个深度学习模型。性能更好的模型将被选用来筛选TCM-WMC 中的肝毒性化合物。

DILI 数据集收集

DILI组合数据集中的化合物是从DILIranks [18], 、LiverTox [19], 、LTKB [20], 、Hepatotox [21]中获取的。DILIranks中的注释通过考虑DILI相关的市场撤回和警告 [18]被分为四种不同的严重程度等级。LiverTox包含关于药物、膳食补充剂和草药引起的肝损伤的全面且基于证据的信息[19]。肝毒性知识库 (LTKB) 包含使用FDA批准的处方药标签评估其在人类中导致DILI潜力的药物 [20]。Hepatotox是一个关于肝毒性药物的数据库, 该文件每年在《临床与生物医学胃肠病学》上发布[21]。 “肝损伤”、“药物引起”是关键词。

在 PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/>)、Nature (<https://www.nature.com/>)、Science Online (<http://www.sciencemag.org/>)、Elsevier Science Direct (<https://www.sciencedirect.com>)、Springer (<https://link.springer.com/>)、Wiley (<https://onlinelibrary.wiley.com/>)、Oxford Academic (<https://academic.oup.com/journals/>) 及其他出版商的数据库中, 以 “肝损伤 (DILI)”、“肝毒性”、“肝脏毒性”、“肝功能衰竭”、“肝损伤”、“肝炎”、“黄疸”、“胆汁淤积”、“肝脏保护”、“护肝”、“护肝作用”、“草药性肝损伤 (HILI)” 为关键词检索相关文献, 并基于 DILI 数据集进行搜索。检索时间限制为 1999–2021 年。不同来源的重复文献和缺乏结构信息的化合物被排除。

人工智能模型构建

化合物的化学结构使用 SMILES (简化分子输入线性表示系统) 编码。使用 PaDEL-Descriptor 软件 [22] 基于 SMILES 字符串计算每个化合物的分子描述符和指纹。通过 PaDEL-Descriptor 软件 (Yap, 2011) 计算了所有化合物的 PaDEL-1D 和 2D 描述符。PaDEL-1D 和 2D 包含 1444 个描述符, 包括原子类型电拓扑状态 (Estate) 描述符、Crippen 的 logP 以及分子线性自由能。

本研究采用了机器学习 (ML) 方法, 包括SGD (随机梯度下降)、kNN (k近邻)、SVM (支持向量机)、NB (朴素贝叶斯)、DT (决策树)、RF (随机森林)、ANN (人工神经网络)、Adaboost、LR (逻辑回归) 来构建肝脏损伤AI模型。本研究还构建了深度信念网络 (DBN) 的两个受限玻尔兹曼机 (RBM)。所有这些AI方法都在相同的数据集上进行训练, 该数据集被随机按约3:1的比例划分为训练集和测试集。用于基于AI方法筛选中药-西药复方中肝毒性化合物的研究流程如图1所示。

模型评估指标的统计

包括分类准确率 (公式 1)、精确率 (公式 2)、召回率 (公式 3)、F1 分数 (公式 4) 以及受试者工作特征曲线 (ROC) 的曲线下面积 (AUC) 在内的五个重要的机器学习方法模型评估指标被用于评估每个模型的性能。其中, AUC 表示 ROC (受试者工作特征) 曲线与坐标轴之间的面积, CA 表示分类准确率, 精确率表示测量值彼此之间的接近程度, 召回率表示召回率。F1 分数的计算公式如公式 (4) 所示。

accounting for 11% in the USA between 1998 and 2013 [3]. The incidence rate of DILI is 23.8 per 100 thousand people in China [4], attracted more research attentions.

The pathogenesis of DILI is not completely clear. Many risk factors lead to DILI, such as the role of host genetic, immunologic, and metabolic factors as well as drug and environmental effects [5, 6]. Although many methods have been developed to predict and assess the risk of DILI, mature and highly accurate methods are still insufficient. Classic methods, including experiments in vitro and in vivo, played a crucial role in predicting DILI. However, drug combinations have become common as multiple diseases coexisting. TCM-WMC have also been increasingly recognized [7–9], covering almost all clinical therapeutic areas in China [10], which increased the complexity of compounds ingested. Hence, it's essential to develop an effective method to screen hepatotoxicity of compounds in TCM-WMC.

With the development of new computing technologies, artificial intelligence (AI) models have been widely used in cheminformatics [11], medical imaging [12], diagnostics [13], bioinformatics [14], and other fields, provided new ideas for screening hepatotoxic compounds. Machine learning and deep learning methods have been increasingly applied to screen hepatotoxic compounds, which treat the high-dimensional chemical structure information as vectors and calculate for prediction or classification purpose in an efficient way [15–17]. In this study, DILI dataset is collected to establish AI models for screening hepatotoxic compounds in TCM-WMC.

Methods

Method design

DILI dataset was collected from public databases and published literatures. Nine machine learning models and a deep learning model were constructed with combined DILI dataset. A better performance model would be chosen to screen hepatotoxic compounds in TCM-WMC.

DILI dataset collection

The compounds in DILI combined dataset were retrieved from the DILIranks [18], LiverTox [19], LTKB [20], Hepatotox [21]. The annotations in DILIranks were assigned four different severity classes by considering DILI-related market withdrawals and warnings [18]. LiverTox contains comprehensive and evidence-based information on drug, dietary supplement, and herbal-induced liver injury [19]. Liver Toxicity Knowledge Base (LTKB) contains drugs whose potential to cause DILI in humans using the FDA-approved prescription drug labels [20]. Hepatotox is a data base on the hepatotoxic drugs file published every year in Gastroent rologie Clinique et Biologique [21]. The keywords of "liver damage", "Drug-induced

liver injury (DILI)", "hepatotoxicity", "liver toxicity", "liver failure", "liver injury", "hepatitis", "jaundice", "cholestasis", "liver protection", "hepatoprotective", "hepatoprotection", "Herb-induced liver injury (HILI)" were searched in PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/>), Nature(<https://www.nature.com/>), Science Online (<http://www.sciencemag.org/>), Elsevier Science Direct (<https://www.Sciencedirect.com>), Springer (<https://link.springer.com/>), Wiley (<https://onlinelibrary.wiley.com/>), Oxford Academic (<https://academic.oup.com/journals/>) and other publishers' databases to search the relevant literatures with DILI dataset. The search time was limited to 1999–2021. Duplicates from different sources and compounds without structures were excluded.

AI model construction

Chemical structures of compounds were coded with SMILES (simplified molecular input line entry system). PaDEL-Descriptor software [22] was used to calculate the molecular descriptor and fingerprint of each compound based on SMILES string. PaDEL-1D and 2D descriptors of all compounds were calculated using PaDEL-Descriptor software (Yap, 2011). PaDEL-1D and 2D contained 1444 descriptors, including atom type electrotopological state (Estate) descriptors, Crippen's logP, and molecular linear free energy.

The machine learning (ML) methods of SGD (Stochastic Gradient Descent), kNN (k-Nearest Neighbor), SVM (Support Vector Machine), NB (Naive Bayes), DT (Decision Tree), RF (Random Forest), ANN (Artificial Neural Network), Adaboost, LR (Logistic Regression) were adopted to build liver injury AI models. Two restricted Boltzmann machines (RBM) of deep belief network (DBN) were also constructed in this research. All these AI methods were trained on the same dataset, which was randomly divided into training set and test set at a ratio of 3:1 approximately. The workflow for the study of screening hepatotoxic compounds in TCM-WMC based AI methods was showed in Fig. 1.

Statistics for model evaluation measures

Five important model evaluation measures for ML methods, including classification accuracy (Eq. 1), Precision (Eq. 2), Recall (Eq. 3), F1 score (Eq. 4), and area under the curve (AUC) of receiver-operating characteristic (ROC) were applied to assess the performance of each model. Therein, AUC represented the area under the ROC (Receiver operating characteristic) curve and the coordinate axis, CA represented the classification accuracy, Precision was how close the measured values that were to each other, Recall represented the recall rate. And the calculation formula of F1 score was as the Eqs. (4).

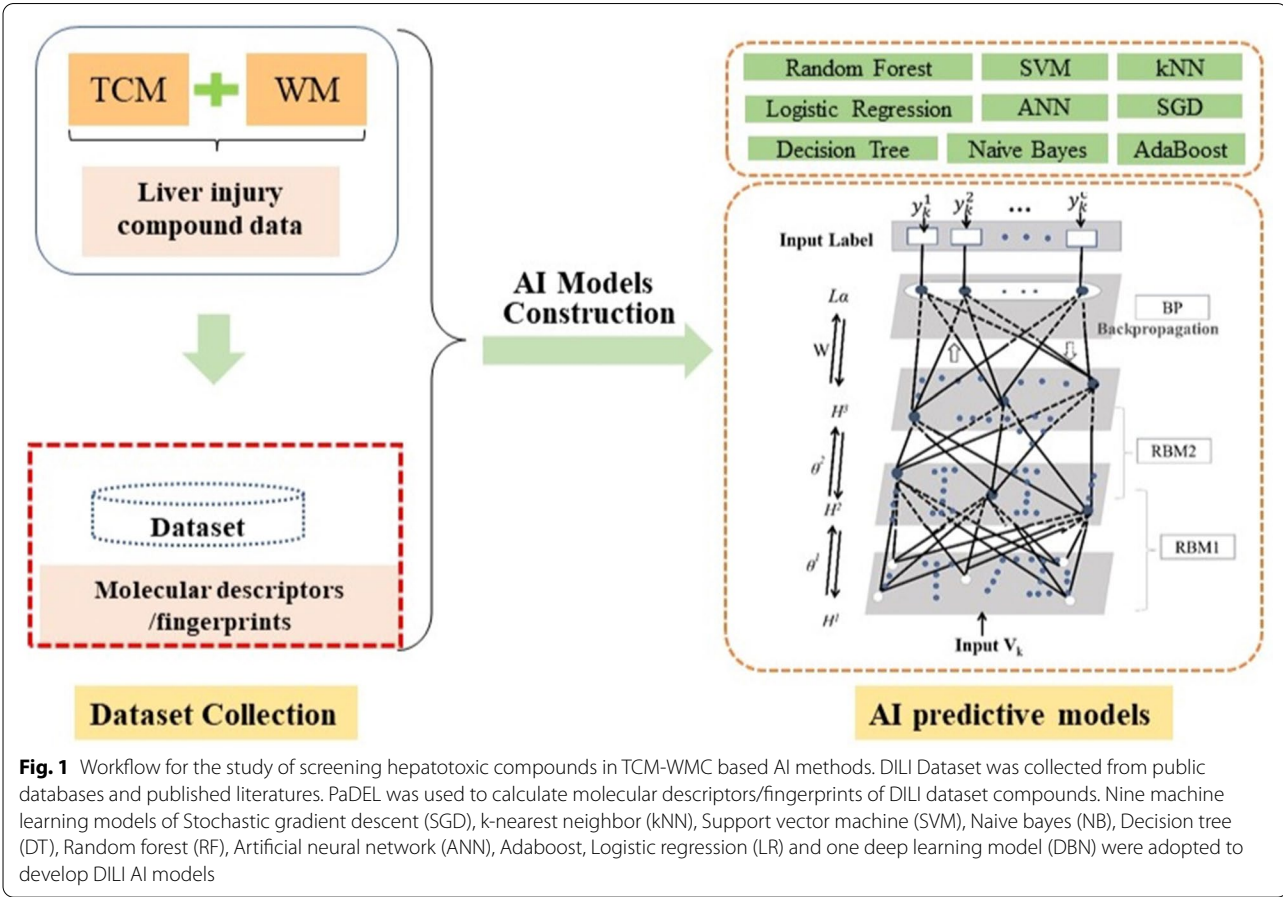
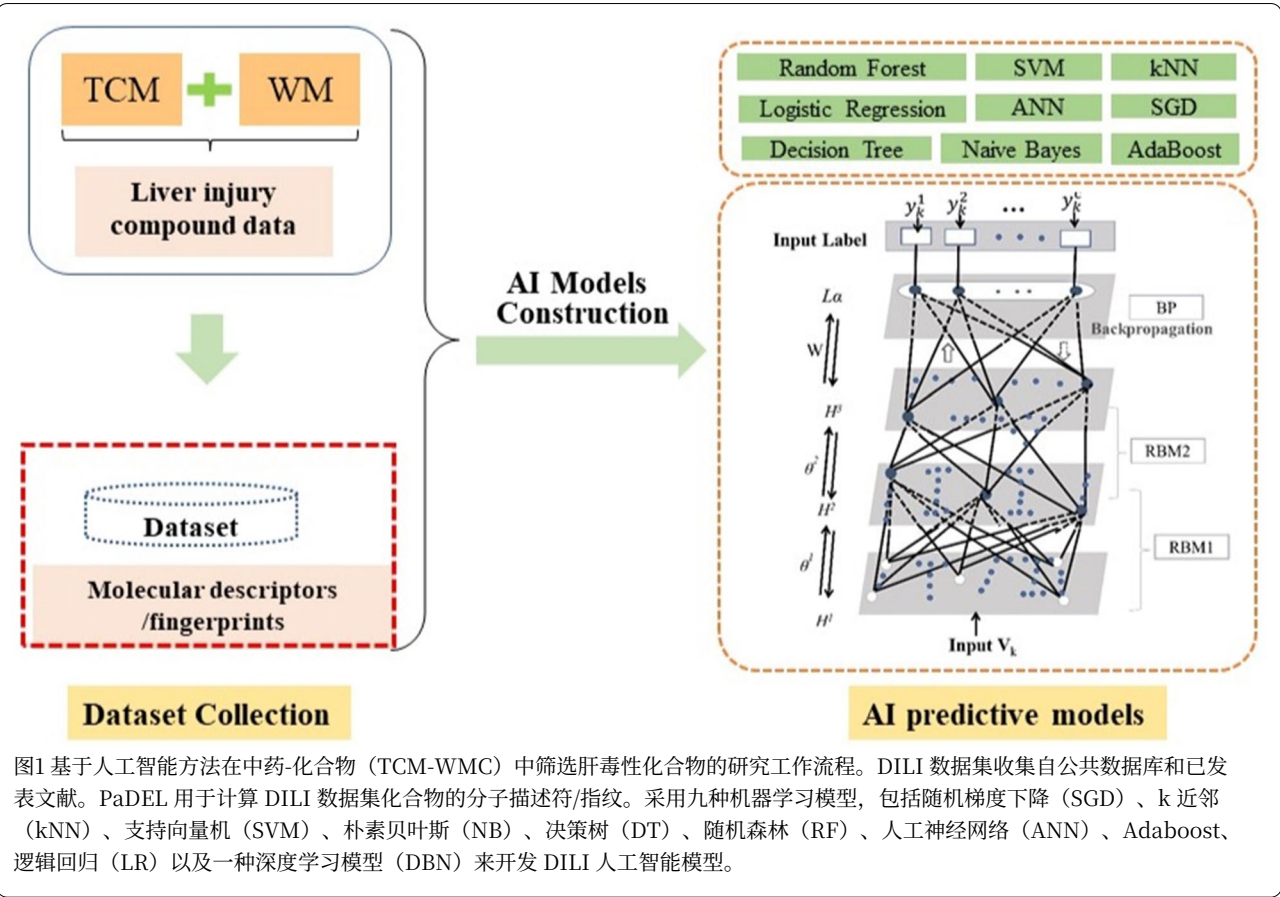


图1 基于人工智能方法在中药-化合物 (TCM-WMC) 中筛选肝毒性化合物的研究工作流程。DILI 数据集收集自公共数据库和已发表文献。PaDEL 用于计算 DILI 数据集化合物的分子描述符/指纹。采用九种机器学习模型, 包括随机梯度下降 (SGD)、k 近邻 (kNN)、支持向量机 (SVM)、朴素贝叶斯 (NB)、决策树 (DT)、随机森林 (RF)、人工神经网络 (ANN)、Adaboost、逻辑回归 (LR) 以及一种深度学习模型 (DBN) 来开发 DILI 人工智能模型。

Fig. 1 Workflow for the study of screening hepatotoxic compounds in TCM-WMC based AI methods. DILI Dataset was collected from public databases and published literatures. PaDEL was used to calculate molecular descriptors/fingerprints of DILI dataset compounds. Nine machine learning models of Stochastic gradient descent (SGD), k-nearest neighbor (kNN), Support vector machine (SVM), Naive bayes (NB), Decision tree (DT), Random forest (RF), Artificial neural network (ANN), Adaboost, Logistic regression (LR) and one deep learning model (DBN) were adopted to develop DILI AI models

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \quad (1) \\ \text{Precision} &= \frac{TP}{TP + FP} \quad (2) \\ \text{Sensitivity} = \text{Recall} &= \frac{TP}{TP + FN} \quad (3) \\ F1 &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4) \end{aligned}$$

结果
从4个与DILI相关的数据库中获得了2875种化合物, 从已发表的与DILI相关文献中的11个数据集中获得了6067种化合物 [23–33]。在排除重复化合物后, 得到了2365种化合物。随后, 还排除了254种在DILIranks中被标注为“DILI关注不确定”的药物。在检查其结构信息后, 没有结构信息的药物也被排除。

结构信息。最终, 本研究共收集了2035种肝损伤化合物, 如图2所示。

在总数据集中, 1505个化合物作为训练集, 530个化合物作为测试集。训练集中肝损伤和非肝损伤化合物的数量比为1125:400, 测试集中肝损伤和非肝损伤化合物的数量比为380:130。

如表1所列, 各模型在训练集上的CA范围为0.686到0.838, F1范围为0.671到0.827, 精确率 (Precision) 范围为0.685到0.832, 召回率 (Recall) 范围为0.686到0.838, AUC范围为0.621到0.814 (受试者工作特征曲线, ROC)。各模型在测试集上的CA范围为0.675到0.767, F1范围为0.636到0.731, 精确率范围为0.555到0.739, 召回率范围为0.675到0.767, AUC范围为0.544到0.739 (ROC)。

上述模型结果显示, RF 在训练集和测试集上都能获得比其他机器学习方法更好的结果 (图 3 和图 4)。DBN 在测试集上获得了 82.2% 的准确率, 其中隐藏层数为 100, 批量大小为 25, 且

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \quad (1) \\ \text{Precision} &= \frac{TP}{TP + FP} \quad (2) \\ \text{Sensitivity} = \text{Recall} &= \frac{TP}{TP + FN} \quad (3) \\ F1 &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4) \end{aligned}$$

Results
Two thousand eight hundred and seventy-five compounds were obtained from 4 DILI-related databases, and 6067 compounds were from 11 datasets in published DILI-related literatures [23–33]. After excluding the duplicate compounds, 2365 compounds was obtained. Subsequently, 254 drugs were also excluded annotated as “Ambiguous DILI concern” in DILIranks. Drugs without structure information were excluded after checking their

structure information. At last, a total of 2035 liver injury compounds were collected in this research, as shown in Fig. 2.

In the total dataset, 1505 compounds were as the training set and 530 compounds were as the test set. The ratio of the number of liver injury and non-liver injury compounds in the training set was 1125:400, and the number ratio of liver injury and non-liver injury compounds in the test set was 380:130, respectively.

As listed in Table 1, CA ranged from 0.686 to 0.838, F1 ranged from 0.671 to 0.827, Precision ranged from 0.685 to 0.832, Recall ranged from 0.686 and 0.838, AUC ranged from 0.621 to 0.814 (Receiver-operating characteristic curve, ROC) for each model on the training set. CA ranged from 0.675 to 0.767, F1 ranged from 0.636 to 0.731, Precision ranged from 0.555 to 0.739, Recall ranged from 0.675 and 0.767, AUC ranged from 0.544 to 0.739 (ROC) for each model on the test set.

Above model results showed that RF could achieved the best results than other machine learning methods on both the training set and the test set (Figs. 3 and 4). DBN obtained 82.2% accuracy on test set, which the number of hidden layers was 100, the batchsize was 25, and the

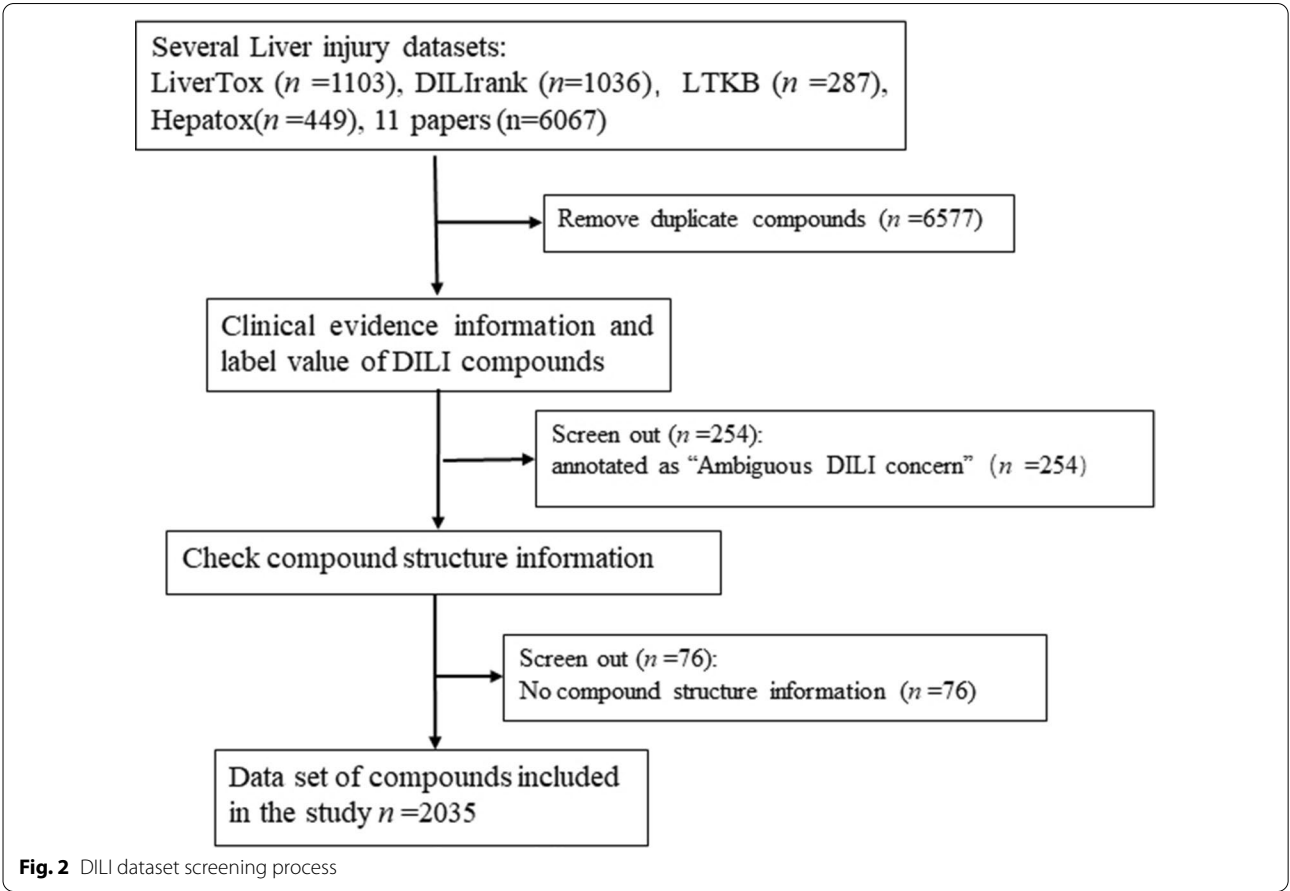
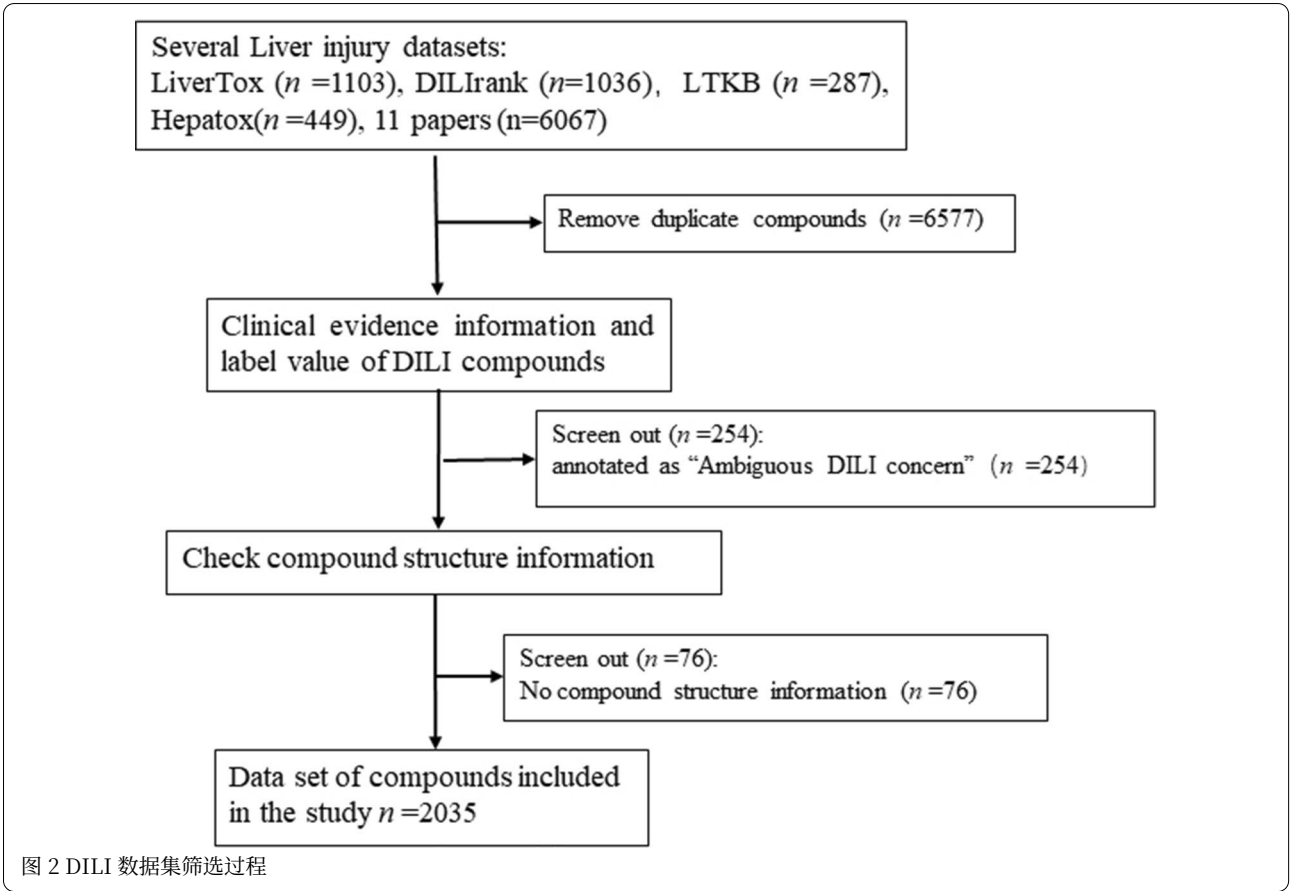


表 1 含药物的肝损伤人工智能模型

数据集	机器学习方法	AUC	CA	F1	精确	召回
训练集	SGD	0.647	0.709	0.715	0.723	0.709
	KNN	0.654	0.722	0.698	0.690	0.722
	SVM	0.785	0.795	0.791	0.788	0.795
	DT	0.710	0.756	0.760	0.764	0.756
	RF	0.814	0.838	0.827	0.832	0.838
	自适应增强	0.785	0.792	0.788	0.785	0.792
	ANN	0.621	0.737	0.671	0.685	0.737
	LR	0.746	0.776	0.757	0.757	0.761
	NB	0.632	0.686	0.694	0.711	0.686
	SGD	0.627	0.682	0.694	0.712	0.682
测试集	KNN	0.574	0.745	0.636	0.555	0.745
	SVM	0.669	0.747	0.712	0.710	0.747
	DT	0.544	0.680	0.679	0.678	0.680
	RF	0.739	0.767	0.731	0.739	0.767
	自适应增强	0.614	0.708	0.707	0.707	0.708
	ANN	0.647	0.694	0.696	0.697	0.694
	LR	0.656	0.733	0.694	0.688	0.733
	NB	0.598	0.675	0.648	0.705	0.675

LR 逻辑回归, RF 随机森林, SVM 支持向量机, kNN k近邻, DT 决策树, NB 朴素贝叶斯, ANN 人工神经网络, SG 随机梯度下降

D

Table 1 AI models for drug-included liver injury

Dataset	ML method	AUC	CA	F1	Precision	Recall
Training set	SGD	0.647	0.709	0.715	0.723	0.709
	KNN	0.654	0.722	0.698	0.690	0.722
	SVM	0.785	0.795	0.791	0.788	0.795
	DT	0.710	0.756	0.760	0.764	0.756
	RF	0.814	0.838	0.827	0.832	0.838
	Adaboost	0.785	0.792	0.788	0.785	0.792
	ANN	0.621	0.737	0.671	0.685	0.737
	LR	0.746	0.776	0.757	0.757	0.761
	NB	0.632	0.686	0.694	0.711	0.686
	SGD	0.627	0.682	0.694	0.712	0.682
Test set	KNN	0.574	0.745	0.636	0.555	0.745
	SVM	0.669	0.747	0.712	0.710	0.747
	DT	0.544	0.680	0.679	0.678	0.680
	RF	0.739	0.767	0.731	0.739	0.767
	Adaboost	0.614	0.708	0.707	0.707	0.708
	ANN	0.647	0.694	0.696	0.697	0.694
	LR	0.656	0.733	0.694	0.688	0.733
	NB	0.598	0.675	0.648	0.705	0.675

LR Logistic regression, RF Random forest, SVM Support vector machine, kNN k-nearest neighbor, DT Decision tree, NB Naive bayes, ANN Artificial neural network, SGD Stochastic gradient descent

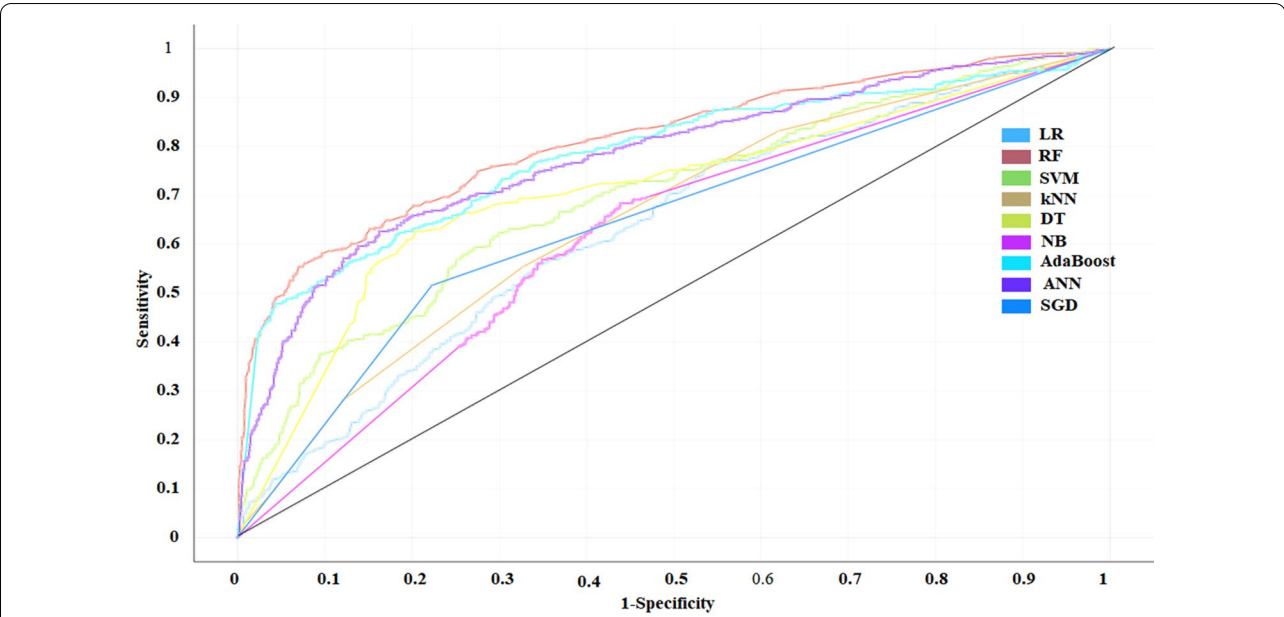


图3 基于人工智能模型的DILI筛选模型训练集的ROC曲线。LR: 逻辑回归, RF: 随机森林, SVM: 支持向量机, kNN: k近邻, DT: 决策树, NB: 朴素贝叶斯, AdaBoost, ANN: 人工神经网络, SGD: 随机梯度下降

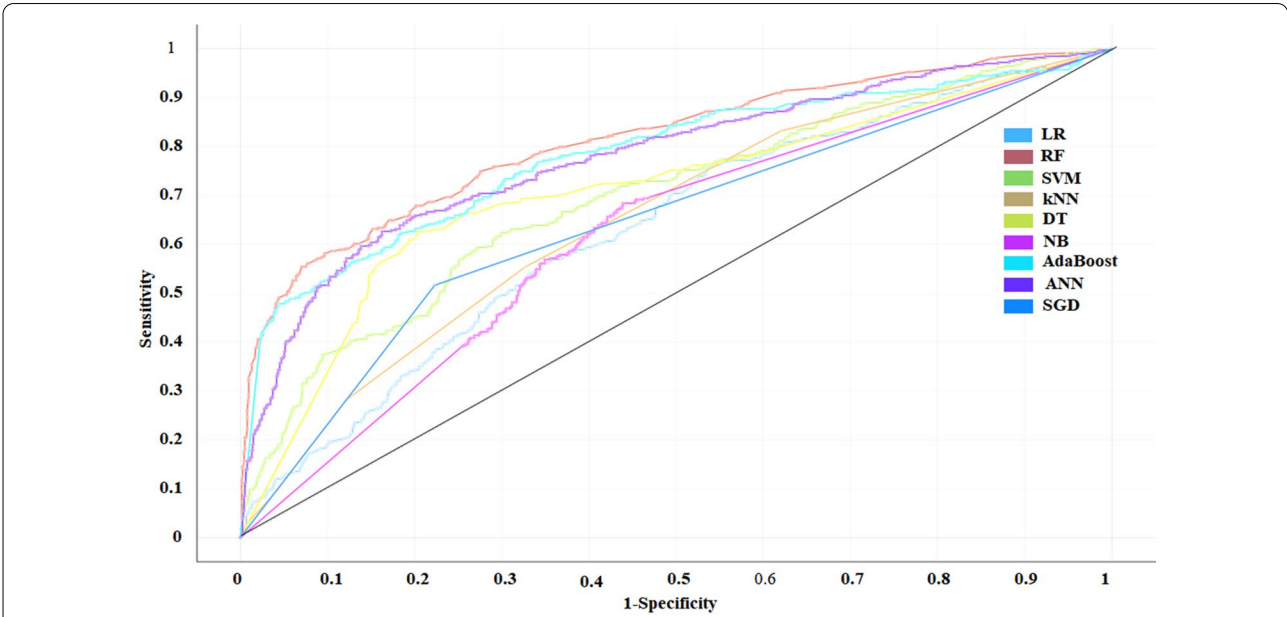


Fig. 3 ROC curve of training set of DILI screening model based on AI model. LR: Logistic regression, RF: Random forest, SVM: Support vector machine, kNN: k-nearest neighbor, DT: Decision tree, NB: Naive bayes, AdaBoost, ANN: Artificial neural network, SGD: Stochastic gradient descent

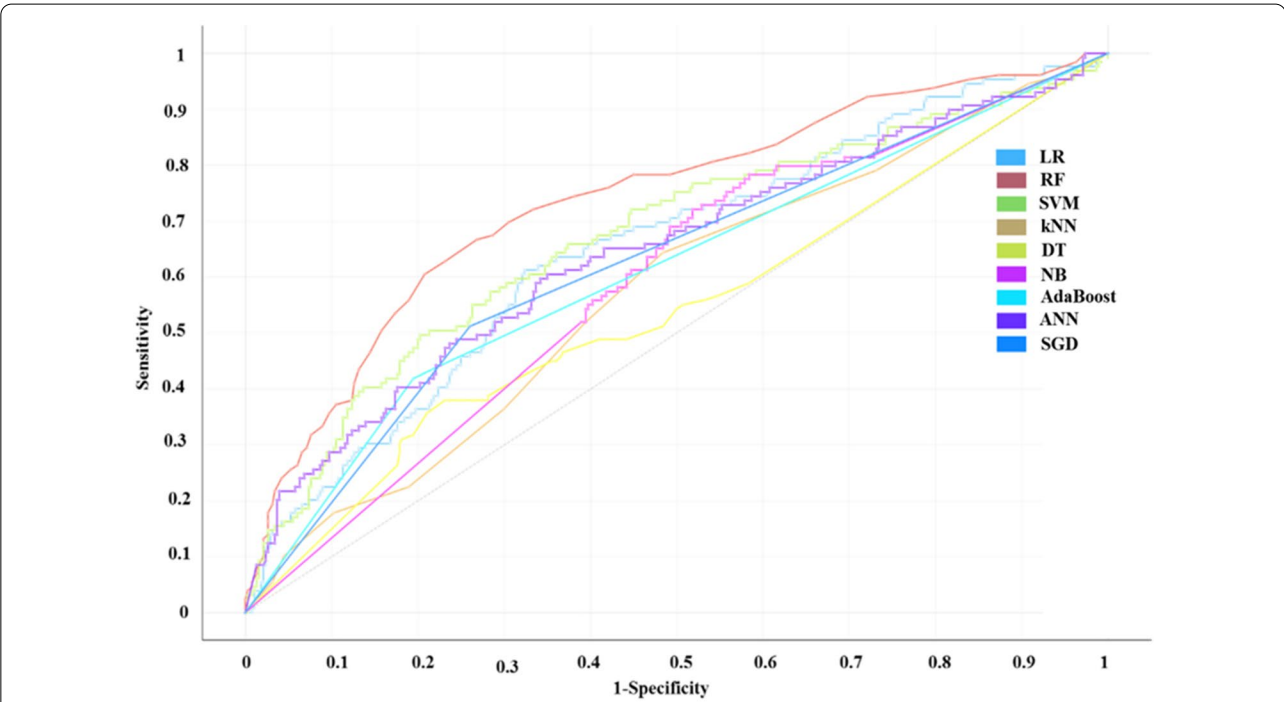


图4 基于AI模型的DILI筛选模型测试集的ROC曲线。LR: 逻辑回归, RF: 随机森林, SVM: 支持向量机, kNN: k近邻, DT: 决策树, NB: 朴素贝叶斯, AdaBoost, ANN: 人工神经网络, SGD: 随机梯度下降

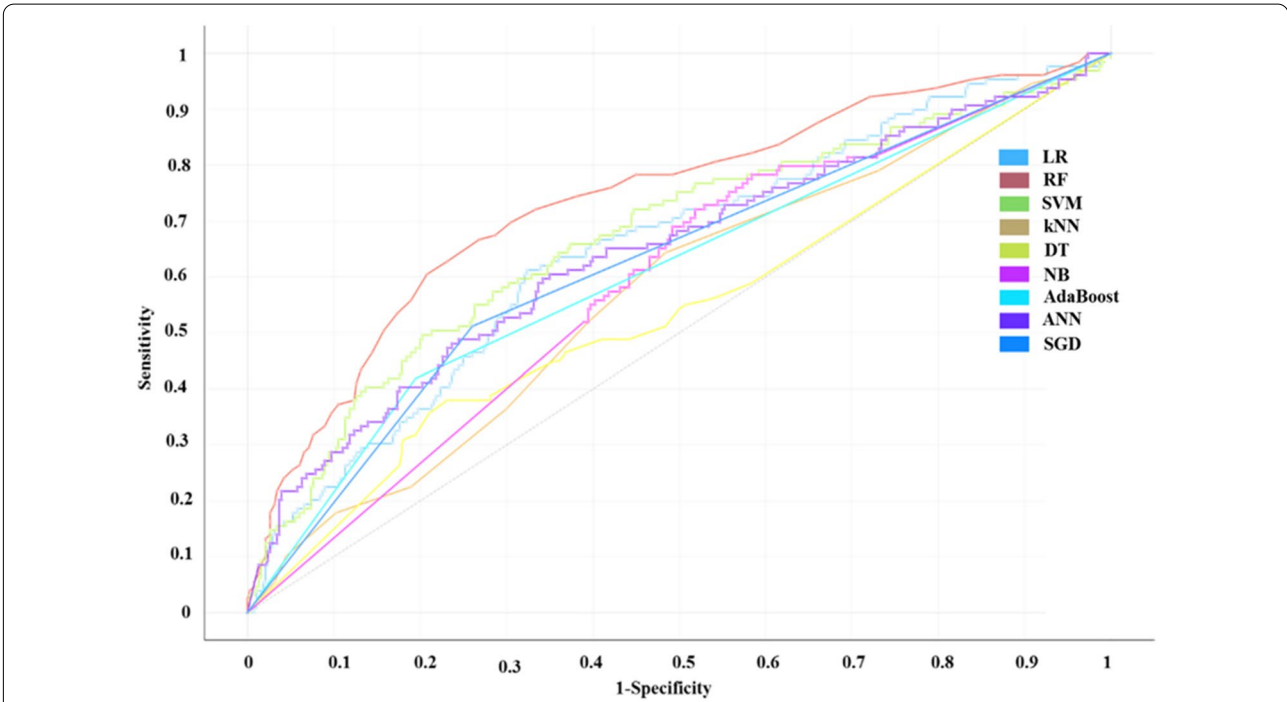


Fig. 4 ROC curve of test set of DILI screening model based on AI model. LR: Logistic regression, RF: Random forest, SVM: Support vector machine, kNN: k-nearest neighbor, DT: Decision tree, NB: Naive bayes, AdaBoost, ANN: Artificial neural network, SGD: Stochastic gradient descent

学习率为1, 迭代次数为400。这一准确率高于测试集中九个机器学习模型的准确率。

讨论
在临床实践中, 药物性肝损伤缺乏特异性和敏感的诊断标准, 诊断的关键部分依赖于

learning rate was 1 with 400 iterations. This accuracy was higher than that of nine machine learning models on the test set.

Discussion
In clinical practice, DILI lacks specific and sensitive diagnostic criteria, and the critical part of diagnosis depends

通过血液检测消除一系列疾病。一旦发生DILI事件, 该药物将被停止用于患者, 甚至可能从市场上撤回。因此, 迫切需要开发一种新的方法来筛查DILI。

当中医与西医结合时, 多种化合物会进入人体。中药化学化合物的复杂性使得中西医结合中的药物性肝损伤极为复杂, 同时也增加了代谢性药物性肝损伤的风险。在中西医结合中, 对具有肝毒性的化合物进行早期预警具有重要的临床用药意义。传统的药物性肝损伤识别依赖于动物实验和临床试验、风险因素评估及病例报告等。这些方法不仅效率低, 而且在人力、物力和财力方面成本高。此外, 实验动物与人类之间可能存在差距, 从而降低肝毒性测试结果在人类中的适用性。相比之下, 最近的药物性肝损伤人工智能模型是基于数学模型构建的。例如, 定量构效关系(QSAR)模型作为代表, 能够在结构信息的基础上推测化合物的具体物理、化学和生物特性。

系统地优化化合物的描述形式及其组合是具有挑战性且至关重要的。SMILES 用一串字符描述三维化学结构。化学结构可以通过一组称为分子指纹或描述符的数值来表征。这些具有高维信息的化合物分子描述符或分子指纹需要通过信息技术(如人工智能)来处理。因此, 在本研究中, 构建了用于筛选中药-魏氏组合(TCM-WMC)中肝毒性化合物的人工智能模型, 使用了9种机器学习模型(SGD、kNN、SVM、NB、DT、RF、ANN、Adaboost、LR)和一种深度学习模型(DBN)。我们发现, DBN模型的性能优于其他9种机器学习方法。

基本的机器学习模型在进行预测或决策时会逐渐变得更好, 但仍然需要一定的指导。更具体地说, 深度学习被认为是机器学习的发展。它使用可编程的神经网络, 使机器能够在无需人类帮助的情况下做出准确的决策。DBN 由多层无监督受限玻尔兹曼机(RBM)网络和一层监督反向传播(BP)网络组成。DBN 的训练过程可以从低层到高层逐步实现, 以训练多层 RBM。每一层 RBM

通过使用前一层的隐藏单元(H)作为输入/可见单元(V)进行训练。药物或化合物的描述符和指纹被用作输入 V_k , 肝损伤的一层监督反向传播(BP)网络的二分类被用作输入标签。无监督RBM和有监督BP的结合实现了对肝毒性化合物的监督预测。在多层RBM训练完成后, 中药-五味子(TCM-WMC)原型化合物/代谢化合物的肝损伤逐渐被分类, 以对TCM-WMC中化合物的肝毒性进行快速预警。

本研究仅关注由单一中西药复方(TCM-WMC)成分引起的肝损伤风险, 并为预测通过中药与西药成分相互作用生成的更多新化合物的肝毒性提供了一条途径。在未来的研究中, 需要进一步阐明中西药复方肝损伤的机制。我们将通过计算更多化合物数据, 进一步关注中西药复方代谢物或复杂中西药复方成分相互作用可能引起的肝损伤风险。总体而言, 本研究将为中西药复方的安全使用提供指导, 并提高临床医生的诊断和治疗能力。

结论

我们比较了不同机器学习和深度学习模型的药物性肝损伤(DILI)模型性能, 发现DBN模型的性能优于其他9种机器学习方法。因此, 这种使用DBN筛选中医药(TCM-WMC)中肝毒性化合物的方法可能有助于指导临床标准和安全用药, 并在临床联合用药中避免肝损伤的风险。

缩写

TCM-WMC: 中西医结合; DILI: 药物性肝损伤; kNN: k近邻; SVM: 支持向量机; NB: 朴素贝叶斯; DT: 决策树; RF: 随机森林; ANN: 人工神经网络; LR: 逻辑回归; DBN: 深度置信网络; CA: 分类准确率; AUC: 曲线下面积; AI: 人工智能; ALF: 急性肝损伤; ML: 机器学习; SGD: 随机梯度下降; ANNs: 人工神经网络; RBM: 受限玻尔兹曼机; ROC: 受试者工作特征曲线; QSAR: 定量构效关系。

致谢

不适用

作者贡献

ZC、CZ 和 HS 构思并设计了实验; ZC、MZ、XZ、YS、HP 和 RQ 收集了数据; ZC、RZ、LY、YJ、TH、XW 和 ZNC 提供了试剂/材料/分析工具/代码; ZC 构建了数据集; ZC 和 MZ 撰写了本文手稿; ZC、MZ、LY、RZ、RQ、CZ 和 HS 修改了手稿。所有作者已阅读并批准最终手稿。

on eliminating a series of diseases through blood testing. Once DILI events occur, the drug will be discontinued for the patient or even withdrawn from the market. Thus, it is urgent to develop a new approach to screen DILI.

Multiple compounds entered into the human body when TCM combined with WM. The complexity of the chemical compounds of TCM makes DILI extremely complicated in TCM-WMC, also increasing the risk of metabolic DILI. Early warning of compounds with hepatotoxicity has a vital clinical medication significance in TCM-WMC. Traditional DILI identification relies on animal experiments and clinical trials, risk factor assessment and case reports, etc. Above methods were not only inefficient but costly in terms of manpower, material, and financial resources. Besides, there may be a gap between experimental animals and mankind, reducing applicability of hepatotoxicity test results in humans. By comparison, recent DILI AI models were constructed based on mathematical models. For instance, quantitative structure–activity relationship (QSAR) model, one representative, was able to speculate on the specific physical, chemical and biological properties of compounds on the basis of structural information of known compounds, thereby achieving qualitative or quantitative screening of unknown compound.

It was challenging and crucial to systematically optimize the description form and their combinations of compounds. SMILES describes a three-dimensional chemical structure with a string of characters. Chemical structures can be characterized by a set of numerical values called molecular fingerprints or descriptors. These compound molecular descriptors or molecular fingerprints with high-dimensional information need to be processed with the application of information technologies, such as AI. Therefore, in this study, AI models was built for screening hepatotoxic compounds in TCM-WMC, with the 9 machine learning models (SGD, kNN, SVM, NB, DT, RF, ANN, Adaboost, LR) and one deep learning model (DBN). We found that DBN model had a better model performance than other 9 machine learning methods.

Basic machine learning models become progressively better at making predictions or decisions, which still need some guidance. More specifically, deep learning is considered to be the evolution of machine learning. It uses a programmable neural network to enable machines to make accurate decisions without humans help. DBN is composed of multi-layer unsupervised restricted Boltzmann machine (RBM) network and one layer supervised back propagation (BP) network.. The training process of DBN can be implemented from low to high level to train multilayer RBMs. Each RBM layer

was trained by using the hidden unit (H) of the previous layer as the input/visible unit (V). The descriptors and fingerprints of drug or compounds were used as input V_k , and the binary classification with one layer supervised back propagation (BP) network of liver injury is used as input label. The combination of unsupervised RBM and supervised BP implements the supervised prediction of hepatotoxic compounds. After the training of the multi-layer RBMs, the liver injury of the prototype compounds/metabolic compounds of TCM-WMC is gradually categorized, to make a rapid warning for hepatotoxicity of compounds in TCM-WMC.

This study only focuses on the risk of liver injury caused by a single compound of TCM-WMC, and provides a path for the prediction of hepatotoxicity of more new compounds formed through the interaction between compounds from TCM and WM. In future research, mechanisms of liver injury in TCM-WMC needs to be further explained. We will further focus on the liver injury possible risk caused by TCM-WMC metabolites or interaction among complex TCM-WMC compounds by calculating more compounds data. Overall, this study will provide guidance for the safe utilization of TCM-WMC and improve the diagnosis and treatment ability of clinicians.

Conclusion

We compared the DILI models’ performance of different machine learning and deep learning models, and found that DBN model had better model performance than other 9 machine learning methods. Therefore, this method of DBN for screening hepatotoxic compounds in TCM-WMC may be helpful to guide the clinical standards and safe medication, and avoid the risk of liver injury in the clinical combination.

Abbreviations

TCM-WMC: Traditional Chinese medicine and western medicine combination; DILI: Drug-induced liver injury; kNN: K-nearest neighbor; SVM: Support vector machine; NB: Naive bayes; DT: Decision tree; RF: Random forest; ANN: Artificial neural network; LR: Logistic regression; DBN: Deep belief network; CA: Classification accuracy; AUC: Area under the curve; AI: Artificial intelligence; ALF: Acute liver injury; ML: Machine learning; SGD: Stochastic gradient descent; ANNs: Artificial neural networks; RBM: Restricted boltzmann machine; ROC: Receiver operating characteristic; QSAR: Quantitative structure–activity relationship.

Acknowledgements

Not applicable.

Author contributions

ZC, CZ and HS conceived and designed the experiments; ZC, MZ, XZ, YS, HP and RQ collected the data; ZC, RZ, LY, YJ, TH, XW and ZNC contributed reagents/materials/analysis tools/codes; ZC constructed the dataset; ZC and MZ wrote this manuscript; ZC, MZ, LY, RZ, RQ, CZ and HS revised the manuscript. All authors read and approved the final manuscript.

资金

本研究得到了中国国家重点研发计划（2019YFC1710400；2019YFC1710405）、国家自然科学基金（编号82104694；编号81803963）、中国博士后科学基金（编号2020M670236；编号2021T140075）的支持。

数据和材料的可用性

本研究中使用的​​所有数据可在请求时从通讯作者处获得。

声明

伦理审批和参与同意 不适用。

同意发表

所有作者都同意该手稿发表在《中国医学》。

利益冲突

作者声明他们没有利益冲突。

作者详情 1

教育部中医内科学重点实验室，北京中医药大学东直门医院，北京，中国。² 复旦大学计算机科学学院，上海，中国。³ 中国中医科学院临床基础研究所，北京，中国。

收到日期：2022年2月14日 接受日期：2022年4月29日 发布日期：2022年5月17日

参考文献。

¹ Chen M 等. FDA 批准药物标签用于药物性肝损伤的研究. Drug Discov Today. 2011;16(15–16):697–703. 2. Stevens JL, Baker TK. 药物安全性测试的未来：扩展视野并聚焦重点. Drug Discov Today. 2009;14(3–4):162–7. 3. Reuben A 等. 1998 至 2013 年成人急性肝功能衰竭的结局. Ann Intern Med. 2016;164(11):724–32. 4. Shen T 等. 中国大陆药物性肝损伤的发生率和病因. Gastroenterology. 2019;156(8):2230–2241.e11. 5. Fontana RJ. 特异性药物性肝损伤的发病机制及临床视角. Gastroenterology. 2014;146(4):914–28. 6. Jay H, Hoofnagle MD, Einar S, Björnsson MD. 药物性肝损伤——类型与表型. N Engl J Med. 2019;381(3):264–73. 7. Zheng R 等. 吴茵颗粒 - 胺碘酮联合治疗心力衰竭合并室性心律失常的临床安全性与疗效：系统综述与荟萃分析. Front Physiol. 2018;9

16. Wang H 等. 从转录组反应预测化学诱导肝毒性终点的深度神经网络模型. 《药理学前沿》. 2019;10:42. 17. Xu Y 等. 药物诱导肝损伤的深度学习. 《化学信息与建模杂志》. 2015;55(10):2085–93. 18. Chen M 等. DILIrank：按人类药物诱导肝损伤风险排序的最大参考药物列表. 《今日药物发现》. 2016;21(4):648–53. 19. Hoofnagle JH 等. LiverTox：关于药物诱导肝损伤的网站. 《肝脏病学》. 2013;57(3):873–4. 20. Thakkar S 等. 肝毒性知识库（LKTb）和药物诱导肝损伤（DILI）分类用于评估人类肝损伤. 《胃肠专家评论》. 2018;12(1):31–8. 21. Quinton A, Latry P, Biour M. Hepatox：肝毒性药物数据库. 《胃肠病学临床与生物学》. 1993;17(5 第2部分):H116–20. 22. Yap CW. PaDEL-descriptor：一个用于计算分子描述符和指纹的开源软件. 《计算化学杂志》. 2011;32(7):1466–74.

出版者注

施普林格·自然在已发布的地图和机构隶属关系中对于管辖权主张保持中立。

Funding

This study was supported by National Key R&D Program of China (2019YFC1710400; 2019YFC1710405); National Natural Science Foundation of China (No.82104694; No.81803963), China Postdoctoral Science Foundation (No. 2020M670236; No. 2021T140075).

Availability of data and materials

All data used in the presented study can get from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All the authors have agreed that the manuscript published in Chinese Medicine.

Competing interests

The authors declare that they have no conflicts of interest.

Author details

¹Key Laboratory of Chinese Internal Medicine of Ministry of Education, Dongzhimen Hospital, Beijing University of Chinese Medicine, Beijing, China. ²School of Computer Science, Fudan University, Shanghai, China. ³Institute of Basic Research in Clinical Medicine, China Academy of Chinese Medical Sciences, Beijing, China.

Received: 14 February 2022 Accepted: 29 April 2022
Published: 17 May 2022

References

1. Chen M, et al. FDA-approved drug labeling for the study of drug-induced liver injury. Drug Discov Today. 2011;16(15–16):697–703. 2. Stevens JL, Baker TK. The future of drug safety testing: expanding the view and narrowing the focus. Drug Discov Today. 2009;14(3–4):162–7. 3. Reuben A, et al. Outcomes in adults with acute liver failure between 1998 and 2013. Ann Intern Med. 2016;164(11):724–32. 4. Shen T, et al. Incidence and etiology of drug-induced liver injury in mainland China. Gastroenterology. 2019;156(8):2230–2241.e11. 5. Fontana RJ. Pathogenesis of idiosyncratic drug-induced liver injury and clinical perspectives. Gastroenterology. 2014;146(4):914–28. 6. Jay H, Hoofnagle MD, Einar S, Björnsson MD. Drug-induced liver injury-types and phenotypes. N Engl J Med. 2019;381(3):264–73. 7. Zheng R, et al. Clinical safety and efficacy of Wenxin Keli-amiodarone combination on heart failure complicated by ventricular arrhythmia: a systematic review and meta-analysis. Front Physiol. 2018;9:487. 8. Han ZZ, et al. Is acupuncture effective in the treatment of COVID-19 related symptoms? Based on bioinformatics/network topology strategy. Brief Bioinform. 2021;22(5):bbab110. 9. Qiu RJ, et al. Outcome reporting from clinical trials of non-valvular atrial fibrillation treated with traditional Chinese medicine or Western medicine: a systematic review. BMJ Open. 2019;9(8):e028803. 10. Zheng R, Chen SQ, Shang HC. The safety evaluation method of herb-drug combination: manifestation-target alliance. Mod Trad Chin Med Mater Med World Sci Technol. 2018;20(10):1734–8. 11. Diaz O, Dalton JAR, Giraldo J. Artificial intelligence: a novel approach for drug discovery. Trends Pharmacol Sci. 2019;40(8):550–1. 12. Bi WL, et al. Artificial intelligence in cancer imaging: clinical challenges and applications. Ca Cancer J Clin. 2019;69(2):127–57. 13. Tomasev N, et al. A clinically applicable approach to continuous prediction of future acute kidney injury. Nature. 2019;572(7767):116–9. 14. Min S, Lee B, Yoon S. Deep learning in bioinformatics. Brief Bioinform. 2017;18(5):851–69. 15. Zhang L, et al. From machine learning to deep learning: progress in machine intelligence for rational drug discovery. Drug Discov Today. 2017;22(11):1680–5.

16. Wang H, et al. Deep neural network models for predicting chemically induced liver toxicity endpoints from transcriptomic responses. Front Pharmacol. 2019;10:42. 17. Xu Y, et al. Deep learning for drug-induced liver injury. J Chem Inf Model. 2015;55(10):2085–93. 18. Chen M, et al. DILIrank: the largest reference drug list ranked by the risk for developing drug-induced liver injury in humans. Drug Discov Today. 2016;21(4):648–53. 19. Hoofnagle JH, et al. LiverTox: a website on drug-induced liver injury. Hepatology. 2013;57(3):873–4. 20. Thakkar S, et al. The Liver Toxicity Knowledge Base (LKTb) and drug-induced liver injury (DILI) classification for assessment of human liver injury. Expert Rev Gastroent. 2018;12(1):31–8. 21. Quinton A, Latry P, Biour M. Hepatox: database on hepatotoxic drugs. Gastroenterol Clin Biol. 1993;17(5 Pt 2):H116–20. 22. Yap CW. PaDEL-descriptor: an open source software to calculate molecular descriptors and fingerprints. J Comput Chem. 2011;32(7):1466–74. 23. O'Brien PJ, et al. High concordance of drug-induced human hepatotoxicity with in vitro cytotoxicity measured in a novel cell-based model using high content screening. Arch Toxicol. 2006;80(9):580–604. 24. Xu JHJ, et al. Cellular imaging predictions of clinical drug-induced liver injury. Toxicol Sci. 2008;105(1):97–105. 25. Greene N, et al. Developing structure-activity relationships for the prediction of hepatotoxicity. Chem Res Toxicol. 2010;23(7):1215–22. 26. Rodgers AD, et al. Modeling liver-related adverse effects of drugs using kNearest neighbor quantitative structure activity relationship method. Chem Res Toxicol. 2010;23(4):724–32. 27. Liew CY, Lim YC, Yap CW. Mixed learning algorithms and features ensemble in hepatotoxicity prediction. J Comput Aided Mol Des. 2011;25(9):855–71. 28. Liu ZC, et al. Translating clinical findings into knowledge in drug safety evaluation—drug induced liver injury prediction system (DILIPs). Plos Comput Biol. 2011;7(12):e1002310. 29. Low Y, et al. Predicting drug-induced hepatotoxicity using QSAR and toxicogenomics approaches. Chem Res Toxicol. 2011;24(8):1251–62. 30. Chen M, et al. The liver toxicity knowledge base: a systems approach to a complex end point. Clin Pharmacol Ther. 2013;93(5):409–12. 31. Zhu X, Kruhlak NL. Construction and analysis of a human hepatotoxicity database suitable for QSAR modeling using post-market safety data. Toxicology. 2014;321:62–72. 32. Huang SH, et al. Developing a QSAR model for hepatotoxicity screening of the active compounds in traditional Chinese medicines. Food Chem Toxicol. 2015;78:71–7. 33. Chen MJ, et al. Quantitative structure-activity relationship models for predicting drug-induced liver injury based on FDA-approved drug labeling annotation and using a large collection of drugs. Toxicol Sci. 2013;136(1):242–9.

Publisher’s Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Ready to submit your research? Choose BMC and benefit from:

- fast, convenient online submission
- thorough peer review by experienced researchers in your field
- rapid publication on acceptance
- support for research data, including large and complex data types
- gold Open Access which fosters wider collaboration and increased citations
- maximum visibility for your research: over 100M website views per year

At BMC, research is always in progress.

Learn more biomedcentral.com/submissions



准备提交您的研究 准备提交您的研究吗？选择BMC并享受以下好处： 选择BMC并享受以下好处：

- 快速、便捷的在线提交
- 经过您在领域有经验的研究人员的彻底同行评审
- 接收后的快速发表
- 对研究数据的支持，包括大型和复杂的数据类型
- 黄金开放获取，促进更广泛的合作和增加引用，为您的研究提供最大的可见性：每年超过1亿次网站浏览量

在BMC，研究工作一直在进行中。

了解更多 biomedcentral.com/submissions